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Hydrocyclones as cell retention devices for an N-1 perfusion bioreactor linked to a continuous-flow stirred tank production bioreactor

Running Title: Hydrocyclone used in linked bioreactor system

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ABSTRACT

A continuous Chinese Hamster Ovary (CHO) cell culture process comprised of a highly proliferative N-1 perfusion bioreactor utilizing a hydrocyclone as a cell retention device linked to a production continuous-flow stirred tank reactor (CSTR) is presented. The overflow stream from the hydrocyclone, which is only partially depleted of cells,

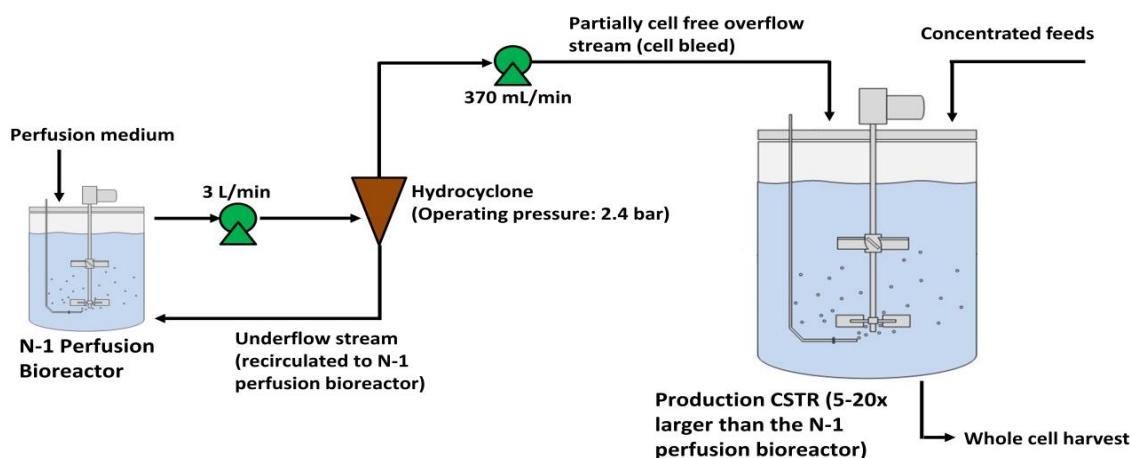
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provides a continuous source of high viability cells from the N-1 perfusion bioreactor to the 5 to 20 times larger CSTR. Under steady-state conditions, this linked-bioreactor system achieved a peak volumetric productivity of 0.96 grams/L/day, two-fold higher than the optimized fed-batch process. The linked bioreactor system using a hydrocyclone was also shown to be 1.8 - 3.1 times more productive than a dual, cascading CSTR system without cell retention.

Graphical Abstract

Hydrocyclones are small and simple devices that alleviate fouling and scale-up concerns associated with conventional perfusion methodologies. We have demonstrated an N-1 perfusion bioreactor utilizing a hydrocyclone for cell retention linked to a 5-20x larger production CSTR operating at steady state. This linked bioreactor system reaches 2x the volumetric productivity of the optimized fed-batch process while enabling the use of legacy manufacturing infrastructure up to 10,000L.



KEYWORDS: Hydrocyclone, perfusion, linked bioreactors, CHO cells, CSTR, continuous-flow stirred tank bioreactor

1. INTRODUCTION

Perfusion with cell retention has been applied to mammalian cell culture for decades. For cells in suspension, a variety of cell retention devices and methodologies have been used, each with its own limitations (Castilho & Medronho, 2002, 2008; Castilho, 2015; Voisard, Meuwly, Ruffieux, Baer, & Kadouri, 2003; Woodside, Bowen, & Piret, 1998). Ultrasonic wave methods suffer from difficulties upon scale up (Castilho & Medronho, 2002) including separation chamber fluid overheating (Pui et al., 1995) and low separation efficiency at high cell densities and perfusion rates (Ryll et al., 2000). Inclined plate cell settlers and other gravity dependent methods also suffer on scale up due to decreased efficiency at high cell densities and the lack of environmental controls (low oxygen, temperature changes) within the device during the long residence times required for high separation efficiencies (Searles, Todd, & Kompala, 1994).

Today, the most commonly used cell retention methods rely on membranes in an external cell retention loop into which the culture is recirculated. While many creative methods minimize or delay fouling, inevitably membranes plug with cellular debris. Such membrane barriers then slowly transform from microporous to ultrafiltration devices, often leading to the undesirable result of product retention and eventually flux decay to zero (Herrero, Prádanos, Calvo, Tejerina, & Hernández, 1997; Kelly & Zydney, 1994; Mercille, Johnson, Lemieux, & Massie, 1994). The high cell densities and low viabilities often associated with highly productive perfusion cultures further accelerate membrane fouling (Karl, Donovan, & Flickinger, 1990; Kelly et al., 2014; Wang et al., 2017). Due

to the issues with design of very large cell retention devices, they are rarely used beyond the 1,000-L volume bioreactor for any long duration process.

Continuous processes operating without cell retention have also been assessed. Originally evaluated decades ago (Hiller, Aeschlimann, Clark, & Blanch, 1991) continuous mammalian cell culture processes operating as simple CSTRs have recently been found capable of delivering volumetric productivities comparable to fed-batch with enough optimization (Khattak, Pferdeort, Brantley, Wiltberger, & Kshirsagar, 2019). Two CSTRs operating in series (or cascade) have also been explored as an option to optimize productivity (Bakker, Schäfer, Beeftink, Tramper, & De Gooijer, 1996).

Hydrocyclones, which have been used for decades in several industries (Bicalho, Mognon, Shimoyama, Ataíde, & Duarte, 2012; Dikov & Sukhanov, 2014; Grommers, Krikken, Bos, & van der Krogt, 2004; Oats, Ozdemir, & Nguyen, 2010; Zemanek, 1973) have also been investigated for their potential as cell retention devices (Deckwer, Medronho, Anspach, & Leubberstedt, 2005; Elsayed, Medronho, Wagner, & Deckwer, 2006; Medronho, Schuetze, & Deckwer, 2005; Ortega-Rivas, 2004; Pinto, Medronho, & Castilho 2008; Schröder, Elsayed, Olownia, & Wagner, 2010). Hydrocyclones generate an internal centrifugal force as fluid circulates through the device at high flowrates, separating cells from bulk culture fluid. Fluid enters through one or two small ports near the top of the device directing the inflow tangential to the inner conical shape. Flow through the chamber creates a primary vortex pushing cells to the wall while the conical taper guides the concentrated cells downward to leave through the bottom port (the underflow). Simultaneously, partially cell-depleted fluid moves upward creating a

secondary vortex until it exits the device through a port at the top (the overflow) (Figure 1) (Castilho & Medronho, 2008; Davailles, Climent, & Bourgeois, 2012).

Unlike membrane-based cell retention methodologies, hydrocyclones generally do not plug with cellular particulates. Shear that occurs as culture fluid passes through the hydrocyclone tends to break apart cell clumps (Pinto et al., 2008). While the flowrates, pressure drops, and shear required for efficient cell separation may promote a single-cell suspension, they may also kill mammalian cells.

Although early work with the hydrocyclones has shown promise with separation efficiencies reaching as high as 99% (Pinto et al., 2008), these experiments were performed at low cell densities that are not representative of current perfusion processes. More recently, viable cell densities up to 50×10^6 cells/mL were achieved in a sustainable perfusion process utilizing a hydrocyclone (Bettinardi, Castan, Medronho, & Castilho, 2020). In experiments performed in our laboratory, compared to membrane cell retention, lower separation efficiencies and shear damage resulting from multiple passes through the hydrocyclone have generally rendered the devices not, or only marginally, suitable for N-1 (seed bioreactor) or production bioreactors. As perfusion rates are increased, more cells are lost to the overflow stream resulting in cellular bleed rates too high for quickly generating the large cell numbers necessary for a production bioreactor inoculation (in the case of N-1 perfusion), or for achieving high cell densities desirable in a continuous perfusion production bioreactor. Also, to achieve high volumetric productivities in continuous perfusion production bioreactors, it can be important to maximize the cell-specific productivity which is usually achieved at lower cellular growth rates (Du et al., 2015; Fussenegger, Mazur, & Bailey, 1997; Gagnon, Nagre,

Wang, & Hiller, 2018). Finally, such low growth rate conditions with low cell turnover usually result in lower cellular viabilities making cells unsuitable for perfusion using hydrocyclones due to the shear forces encountered.

Recent studies in our laboratory have demonstrated that a perfusion N-1 bioreactor using a membrane-based cell retention device linked to a 5-20 times larger CSTR may be as much as twice as productive as a standard fed-batch mode of operation (Gagnon, Nagre, Wang, Coffman, & Hiller, 2019; Hiller, Gagnon, & Coffman, 2019). Additionally, a linked bioreactor system may be more amenable to scale up than a standard perfusion bioreactor since the linked bioreactor system does not require cell retention in the larger production bioreactor. In a linked bioreactor system, the cells in the N-1 culture are typically maintained in a highly proliferative and highly viable state with somewhat high perfusion and cell bleed rates. High viability cells are generally more resistant to hydrodynamic shear and therefore might withstand the hydrocyclone's high shear environment. Low efficiencies of the hydrocyclone result in a considerable number of viable cells entrained in the partially cell-free fluid leaving the device (overflow stream) which can be used as a continuous inoculum source for the production CSTR. The very fast flowrates and processing times mean that a very small hydrocyclone can be used to separate cells from 200-550 liters of cell culture fluid per day. While the fluid dynamics of the fluid flow complicate scale up, it is a trivial matter to simply increase the number of hydrocyclones of an identical geometry operating in parallel (Bettinardi et al., 2020) with a single pump. Hydrocyclone arrays could be constructed from stainless steel and cleaned and sterilized in place on industrial bioreactors (Castilho, 2015) or 3-D printed and sterilized by gamma irradiation or heat and provided as single use components.

Unlike fed-batch bioreactor operations which generate a single large-volume harvest usually after 10-18 days, a linked N-1 perfusion to CSTR system operating at a steady state would generate a continuous flow of product from the bioreactor, likely with a consistent titer and product quality. Such a continuous cell culture process could take advantage of the numerous economic and operational benefits of a continuous purification process (Castilho, 2015; Zydney, 2016).

In the current work, an N-1 perfusion bioreactor using hydrocyclone cell retention achieved steady state conditions with a high cellular growth rate and high cell bleed. The hydrocyclone overflow stream supplied a continuous inoculum to a 5 to 20 times larger production CSTR which did not use cell retention. The process was compared to a dual cascading CSTR system in which the N-1 bioreactor operates as a CSTR without cell retention.

2. MATERIALS AND METHODS

2.1 CELL LINES AND INOCULUM MAINTENANCE

Experiments used a CHO-K1 cell line stably transfected with DNA vectors encoding a recombinant antibody using the glutamine-synthetase (GS) expression system (Lonza, Basel, Switzerland). Using CD CHO AGT™ medium (Thermo Fisher Scientific, Carlsbad, CA), cells were expanded in shake flasks (Corning, Oneonta, NY) maintained in a humidified incubator (Kuhner, Basel, Switzerland) at 36.5°C, 5% carbon dioxide, and 140 rpm agitation rate. Shake flasks were inoculated at 0.3×10^6 and 0.2×10^6 cells/mL for 3- and 4-day passages, respectively.

2.2 3D PRINTED HYDROCYCLONES

The hydrocyclone incorporated a 3 mm underflow diameter, 2 mm overflow diameter, 10 mm swirl chamber diameter, and 71 mm conical length. It was 3D printed with a Form 2 inverted stereolithography printer (Formlabs, Somerville, MA) using biocompatible and autoclavable dental resin (Formlabs, Somerville, MA), and ultraviolet light cured and autoclave sterilized prior to use.

2.3 MEDIA AND FEEDS

All media and feeds were chemically defined, proprietary formulations developed in house. Some details of the formulations are listed in Table 1. Poly-vinyl alcohol was used as a shear protectant.

2.4 GENERAL EQUIPMENT

3 or 15-L glass bioreactors (Applikon, Inc., Schiedam, Netherlands) with 1.4 or 8-L working volumes were equipped with Hamilton EasyFerm Plus pH probes and VisiFerm DO probes (Hamilton, Reno, NV) linked to BioNet controllers (Broadley James, Irvine, CA) which maintained pH and dissolved oxygen. The 3-L bioreactors were agitated with a single Rushton impeller that was 4.5 cm in diameter and the 15-L bioreactor used a lower Rushton impeller (7.6 cm in diameter) and an upper down-pumping pitched-blade impeller (7.6 cm in diameter). In the 15-L bioreactor, dissolved gas control used a dual sparging strategy employing a 100 μ m sintered steel sparger to supply pure oxygen and a 7x1mm drilled hole sparger using air primarily for carbon dioxide removal. In the 3-L bioreactors, a single 7x1mm drilled hole sparger was initially used to supply oxygen to the culture. Once the oxygen demand increased to high levels, a 15 μ m sintered steel sparger was used to supply oxygen while the drilled hole sparger utilized pure oxygen for

both oxygen delivery and carbon dioxide removal. CSTRs were maintained at a constant working volume by a ChemTec pump (Parker Domnik Hunter, Oxnard, California) which removed whole cell culture in response to an increase in bioreactor weight (Sartorius Stedim, Aubagne, France).

2.5 N-1 PERFUSION USING A HYDROCYCLONE AS A CELL RETENTION DEVICE LINKED TO A PRODUCTION CSTR

The 15-L N-1 perfusion bioreactor was inoculated at 1.5×10^6 cells/mL, maintained at 36.5°C, and agitated at 80 W/m³. The dissolved oxygen was maintained at 40% of air saturation and dissolved carbon dioxide between 4 and 10% of saturation. Perfusion using the hydrocyclone began on Day 3. A peristaltic pump (600 series, Watson Marlow, Wilmington, MA) supplied flow to the hydrocyclone at approximately 3 L/min to create a pressure drop of 2.4-2.6 bar within the device. Concentrated cells were recirculated to the bioreactor in an unrestricted underflow stream. This recirculation loop consisted of approximately 15 cm of 25.4 mm inner diameter (ID) tubing which fit over the lower portion of the hydrocyclone. This 25.4 mm ID tubing was reduced to 9.5 mm ID tubing over 4 cm using a 3D printed dual hose barb reducer. The 9.5 mm ID tubing was attached to the reducer and welded onto the bioreactor. A second peristaltic pump (500 series, Watson Marlow, Wilmington, MA) controlled the partially cell-free overflow stream at a flowrate of 370 mL/min for a maximum daily perfusion volume of 532 L. Although the overflow stream was restricted with the peristaltic pump for purposes of accurate control, earlier experiments showed the flowrate of an unrestricted overflow stream would have been nearly identical. A diagram illustrating the linked bioreactor system using a hydrocyclone is shown in Figure 2a.

In order to use a fraction of the maximum perfusion capacity, the perfusion rate was controlled by restricting the hydrocyclone operational time. For the N-1 perfusion in this experiment, the hydrocyclone was operated for 30 minutes per day or less. Power outlet timers turned the inlet and overflow pumps on and off every 3 hours and the required hydrocyclone operation time was distributed evenly over 8 intervals per day. Once the pumps turned on, the hydrocyclone system reached hydrodynamic equilibrium within a few seconds and startup time was negligible. Between power cycles, fresh perfusion medium was added through connections just prior to the inlet pump and immediately after the hydrocyclone in the underflow stream. This practice allowed most of the cells in the recirculation loop to be washed back into the bioreactor between power cycles.

At the manufacturing scale, it is intended that the entire hydrocyclone overflow stream will provide fresh cells to an N-stage production bioreactor that is between 5 and 20 times the volume of the N-1. In our experimental setting, the N-1 bioreactor was larger than the production CSTR, therefore the volume ratios were simulated by adding only the appropriate volume of overflow stream to the production CSTRs while diverting the remaining overflow to waste. On Day 4, the overflow stream began to supply cells to three different N-stage production CSTRs simulating N-1 to N volume ratios of 1:5, 1:10, and 1:20. In the envisioned large-scale process, cell addition to the production CSTRs would ideally begin as soon as perfusion is started. In this experiment, the cell addition started approximately 16 hours after perfusion start so the initial automated operations could be closely monitored.

The production CSTRs initially contained basal medium at 70% of the final working volume and the overflow stream gradually supplied cells to the CSTRs using peristaltic

pumps (300 series, Watson Marlow, Wilmington, MA) operating on a similar intermittent operating schedule as the hydrocyclone inlet and overflow pumps. To ensure that the cells added to the production CSTRs were fresh and represented the steady-state operational separation of the system, the pumps supplying cells to the CSTRs were timed to start after one minute of hydrocyclone operation (during which the entire overflow stream was directed to waste). Once the working volume in the CSTRs accumulated to 1.4 L, the bioreactor working volume was maintained constant by continuously removing whole cell harvest from the bioreactor. The production CSTRs were maintained at the same environmental conditions (dissolved gases, pH, temperature, agitation) as the N-1 perfusion bioreactor. Nutrients were supplied directly to the CSTRs using highly concentrated nutrient feeds to maintain the total residual amino acid levels above approximately 40 mM (measured by off-line analysis), ensuring that no individual amino acid was completely depleted.

The N-1 perfusion rate was increased daily on Days 3-17 to maintain a target cell specific perfusion rate (CSPR) of ~50 picoliters per cell per day (pL/cell/day). On Days 17-23, a perfusion rate of 1.05 reactor volumes per day (RV/day) was maintained to allow the linked bioreactor systems to reach steady-state conditions. Following the first steady state, the perfusion rate was gradually increased to 1.40 RV/day and held constant from Day 25-33 to attain a second steady state.

To control lactate accumulation below a growth inhibitory level in the N-1 perfusion and production CSTRs, a glucose limiting technique called HIGH-end pH Delivery of Glucose (HIPDOG) (Gagnon et al., 2011) was implemented to feed glucose based on pH. The pH setpoint during HIPDOG control was 7.075 ± 0.05 , controlled by a 1 M

sodium/potassium carbonate solution (0.94 M sodium carbonate and 0.06 M potassium carbonate) on the low end and a 500 g/L glucose solution on the high end. In the N-1 perfusion, lactate is flushed from the bioreactor as a result of the perfusion process itself. Consequently, lactate exhaustion could cause HIPDOG control to fail because without some residual lactate, presumably there would not be a pH response to signal glucose depletion (Hiller, Gagnon, & Ovalle, 2018; Hiller, Ovalle, Gagnon, Curran, & Wang, 2017). To prevent HIPDOG failure, 1 M sodium/potassium carbonate was added to the bioreactor to generate an upward driving force on pH to ensure that lactate would not be fully exhausted. The carbonate solution was added at a rate of less than 1% of the perfusion volume and adjusted to maintain the lactate concentration in the bioreactor between 0.5 and 2 g/L. The CSTR cultures were initially in a highly proliferative state with the corresponding tendency to produce high levels of lactic acid. During this initial growth phase, HIPDOG was implemented to control lactate accumulation; once the growth rate slowed and the rate of lactic acid production was anticipated to have dropped (Hartley, Walker, Chung, & Morten, 2018; Mulukutla, Gramer, & Hu, 2012), HIPDOG control was discontinued. Following HIPDOG control, the pH setpoint was 7.05 ± 0.10 , controlled with 1 M sodium/potassium carbonate on the low end and carbon dioxide on the high end. The glucose feed rates post HIPDOG were determined using historical glucose consumption rates and adjusted based upon measured residual glucose levels.

2.6 CASCADING CSTRS

To demonstrate the value of the hydrocyclone as a cell retention device, we evaluated a simpler alternative not requiring perfusion. It was contemplated that the N-1 bioreactor could be operated as a CSTR to add the same volume and quantity of cells to the

production CSTRs if the volume of the N-1 bioreactor was doubled and culture conditions maintaining a high growth rate were utilized. Because we predicted that an N-1 CSTR would not achieve as high cell densities as the N-1 perfusion bioreactor using the hydrocyclone, in this experiment the simulated volume ratios of the N-1 CSTR to the production CSTRs were halved, thereby simulating volume ratios of 1:2.5, 1:5, and 1:10. All other experimental conditions were identical to the experiment using the hydrocyclone (media, dissolved gas control, temperature, agitation, pH). A diagram of the cascading CSTRs process is shown in Figure 2b.

The N-1 CSTR was inoculated at 0.8×10^6 cells/mL with the same CHO cell line used in the hydrocyclone linked bioreactor experiment. Dilution began on Day 3 at 0.25 RV/day by adding N-1 perfusion medium (Table 1). The dilution rate was increased to 0.54 RV/day on Day 4 and held constant for the remainder of the experiment. The perfusion/dilution medium was added semi-continuously to the N-1 bioreactor in equivalent doses every 15 minutes.

Like the hydrocyclone linked bioreactor system, it is intended that the entire cell bleed from the N-1 CSTR will feed a production CSTR between 2.5 and 10 times larger in volume in a manufacturing setting. In this experimental setup, the volume ratios were simulated by adding the appropriate volume of cell bleed to the production CSTRs while discarding the excess. The production CSTRs were initiated and operated similarly to the hydrocyclone linked experiment (70% partial volume start and continuous level control once at working volume).

Lactate accumulation was controlled in the dual cascading CSTR system in an identical manner as previously described for the hydrocyclone linked bioreactors. Also using a similar approach as the hydrocyclone linked bioreactors, a concentrated nutrient feed (Table 1) was delivered directly to the production CSTRs to maintain the residual amino acid concentration above approximately 40 mM.

On Day 35, after the production CSTRs simulating 1:2.5 and 1:5 volume ratios maintained steady state conditions for several days, the cell bleed addition from the N-1 CSTR was stopped to evaluate the production CSTRs as independent unit operations. The dilution rates were maintained constant by replacing the cell bleed volume from the N-1 with fresh perfusion/dilution medium.

2.7 ANALYTICAL METHODS

Analytical methods for viable cell density, cell viability, metabolites, amino acids, offline pH, offline CO₂, and antibody product concentration were identical to those described in a previous publication (Hiller et al., 2017).

3 RESULTS AND DISCUSSION

3.1 N-1 PERFUSION USING A HYDROCYCLONE AS A CELL RETENTION DEVICE LINKED TO A PRODUCTION CSTR

On Days 3 to 6, the hydrocyclone was initially operated at a pressure of 2.6 bar; however, during this time, the growth rate dropped significantly, and the cell viability began to trend downward. In attempt to reduce shear and recover the culture, the operating pressure was decreased to 2.4 bar on Day 6. Although the separation efficiency decreased slightly, the culture quickly returned to near exponential growth and the viability

stabilized. Operating at 2.4 bar, 92-96% of cells entering the hydrocyclone were recovered in the underflow stream returning to the bioreactor (Figure 3a). This percent recovery or total separation efficiency (E_t) (Castilho & Medronho, 2008) was determined using the following equation:

$$E_t = \left(1 - \frac{X_o Q_o}{X_{brx} Q}\right) * 100\%$$

where X_o and X_{brx} denote the viable cell density of the overflow stream and bioreactor, respectively, and Q_o and Q denote the flowrates of the overflow stream and inlet stream, respectively. Reduced or centrifugal separation efficiency (E') can also be calculated using the equation below (Castilho & Medronho, 2008):

$$E' = \left(1 - \frac{X_o}{X_{brx}}\right) * 100\%$$

In this experiment, an average reduced separation efficiency of 54% was observed.

Since the hydrocyclone did not provide perfect separation, as the perfusion rate increased, the effective cell bleed from the overflow stream also increased (Figure 3b). The effective bleed rate is defined as the rate of cell bleed that would occur if it was being removed directly from the bioreactor on a continuous basis. This cell bleed rate is then equivalent to that typically defined in perfusion bioreactor operation when the cell retention system is 100% effective (as a membrane cell retention system would be) and it is necessary to remove cells directly from the culture in order to achieve a steady-state (biomass constant) condition.

The N-1 perfusion reached the first steady state with respect to most parameters on Days 18 - 23 at a perfusion rate of 1.05 RV/day, effective cell bleed rate of 0.54 RV/day, and an average viable cell density of 20.4×10^6 cells/mL (Figure 4a). The linked production CSTRs also reached steady state conditions (Table 2) at viable cell densities ranging from $19.0 - 23.5 \times 10^6$ cells/mL (Figure 4b, 4c, and 4d) and dilution rates between 0.10 and 0.27 RV/day (Figure 5a). A second steady state was achieved in the N-1 perfusion on Days 27-33 at a perfusion rate of 1.40 RV/day, effective cell bleed rate of 0.70 RV/day, and an average viable cell density of 16.5×10^6 cells/mL. The linked production CSTRs reached a second steady state (Table 2) at viable cell densities between 11.5 and 15.0×10^6 cells/mL and dilution rates ranging from 0.10 to 0.32 RV/day. At steady state timepoints, the N-1 perfusion sustained a cell viability of 91-94% while the production CSTRs maintained cell viabilities of 93-99%. The lower viability in the N-1 bioreactor is likely due to the high shear environment created by the hydrocyclone. The cell densities achieved in the N-1 perfusion bioreactor in the experiments presented here were somewhat lower than those achieved by other researchers performing perfusion with CHO cells using hydrocyclones as cell retention devices (Bettinardi et al., 2020).

In the N-1 perfusion bioreactor, HIPDOG control was initiated on Day 2 and continued for the duration of the experiment. At perfusion rates higher than 1.2 RV/day, the perfusion medium supplied sufficient glucose to the culture without additional supplementation by HIPDOG control. HIPDOG control was implemented to maintain lactic acid concentrations at non-inhibitory levels from Days 10-18, Days 11-18, and Days 13-18 for the 1:5, 1:10, and 1:20 volume ratio CSTRs respectively. Glucose and lactate profiles are shown in Figures 5b and 5c, respectively. Dissolved carbon dioxide

and osmolality were maintained within 4-10% of saturation and 280-380 mOsm/kg, respectively. Ammonium levels remained below 6 mM in the N-1 perfusion and the 1:5 production CSTR at the steady state timepoints. The 1:10 and 1:20 production CSTRs (operating at low dilution rates) reached slightly higher ammonium levels during Steady State 2 at 7.5mM and 9.6mM, respectively.

On Day 34, there was a perturbation in the N-1 perfusion bioreactor when the volume dropped to approximately 75% of the setpoint likely due to a malfunctioning media pump. During this low volume period, the top impeller was partially exposed and potentially caused excess shear at the liquid/air interface. In an initial attempt to recover the culture, the working volume was restored by delivering additional perfusion media and the perfusion rate was maintained at 1.40 RV/day. However, lactate spiked to inhibitory levels as high as 3.2 g/L while the cell density and viability dropped to 4.3×10^6 cells/mL and 86%, respectively. Using a different approach to recover the culture, on Day 38 the perfusion rate was dropped to 0.4 RV/day to induce lactic acid consumption and reduce loss of cells to the hydrocyclone overflow stream. When the growth rate and viability improved, the overall perfusion rate was slowly increased while maintaining a CSPR of ~ 50 pL/cell/day. On Day 49, the culture had fully recovered to the conditions achieved in the first steady state and conditions maintained reasonably stable until the end of the experiment on Day 56. In retrospect, using a perfusion medium with a lower glucose concentration would have provided more flexibility to control lactate and expedite culture recovery.

The steady state volumetric productivities of the N-1 perfusion and the production CSTRs were calculated using two methods. In the first method, the antibody product

concentration (Figure 5d) on the last day of the steady state was multiplied by the dilution or perfusion rate. However, this method would only be accurate if the product concentration was constant and the low dilution rates in the production CSTRs significantly delayed this response. The second and more accurate method of calculating steady state volumetric productivity utilized a material balance on product produced by the linked bioreactors. This calculation considered product entering the production bioreactor from the overflow stream, product exiting in the harvest stream, and product accumulating in the bioreactor. The total product produced was plotted against time (Figure 5e) and a linear regression was performed to determine the volumetric productivity.

The linked bioreactor system had a volumetric productivity of 0.84 - 0.96 g/L/day during Steady State 1 and 0.54 - 0.55 g/L/day during Steady State 2. Due to the moderate genetic instability of this CHO cell line (Gagnon et al., 2019; Hiller et al., 2019), Steady State 2 volumetric productivities are likely slight underestimates. With only minimal process development, at Steady State 1 this continuous linked bioreactor system is 1.8 - 2 times more productive than the optimized 12-day fed-batch process which has a cumulative volumetric productivity of 0.48 g/L/day, and is also significantly more productive than the N-1 perfusion bioreactor operating independently (0.54 g/L/day, Table 2). Finally, the results of the linked system using the hydrocyclone compare very favorably with the productivity previously obtained for the same cell line, media, and environmental parameters but using a membrane as the cell retention device (Gagnon et al., 2019; Hiller et al., 2019). With the Steady State 1 operating conditions, a single hydrocyclone of the design used in the experiments presented here could operate as the cell retention device

for a 500-L N-1 perfusion bioreactor with the overflow stream feeding a highly productive production bioreactor between 2,500-10,000 liters without concerns of fouling. As previously shown (Gagnon et al., 2019; Hiller et al., 2019), a linked bioreactor process could potentially operate indefinitely at a highly productive steady state by occasionally (depending upon the genetic stability of the cell line) re-starting the N-1 bioreactor with cells of a lower generational number.

The media requirements for this linked bioreactor system are low and would result in a highly efficient and practical manufacturing process (Table 2). In this system the N-1 perfusion bioreactor generates no waste because the entire overflow stream is used to provide cells to the CSTR production bioreactors. The linked bioreactor system would require a total of 0.10 - 0.32 RV/day (production reactor volume) of media in comparison to the 1.05 - 1.40 RV/day the N-1 perfusion bioreactor requires as a standalone unit operation. While we believe the comparisons of the linked bioreactor system to the standalone perfusion bioreactor from these experiments are useful, it is understood that the perfusion bioreactor conditions were not optimized for standalone operation.

3.2 CASCADING CSTRS

On Days 0-2, a significant temperature probe miscalibration caused the N-1 CSTR to be maintained $\sim 3^{\circ}\text{C}$ lower than normal, resulting in a lag in growth. The temperature was corrected on Day 2 and the culture quickly returned to exponential growth. HIPDOG control was successful in controlling lactate accumulation in the N-1 CSTR starting on Day 7 and continuing until the end of the experiment. HIPDOG was used on Days 5-17, Days 6-17, and Days 11-20 for the 1:2.5, 1:5, and 1:10 simulated volume ratio production

CSTRs, respectively. Dissolved carbon dioxide and osmolality were maintained in suitable physiological ranges, 4-10% of saturation and 280-400 mOsm/kg, respectively.

The N-1 CSTR, operating at a dilution rate of 0.53 RV/day, reached a steady state on Day 14 at 10×10^6 viable cells/mL (Figure 6a) which was maintained until the conclusion of the experiment on Day 43. The linked production CSTRs also reached steady state conditions which are summarized in Table 3. In the production CSTRs there was a significant overshoot in viable cell density during the initial startup phase prior to reaching steady state conditions. To avoid this overshoot, the dilution rate in the production CSTR could initially be set high and then lowered as the growth rate slows. However, this may not improve the overall volumetric productivity of the system. At the steady state timepoints, the N-1 CSTR and the production CSTRs simulating 1:2.5 and 1:5 volume ratios sustained cell viabilities exceeding 97% (Figures 6b and 6c). The viability in the production CSTR simulating a 1:10 volume ratio had a lower steady state viability of 83-87% (Figure 6d), potentially due to the accumulation of certain inhibitory metabolites at this low dilution rate (Figure 7a) seen previously for this cell line (Gagnon et al., 2019; Mulukutla, Kale, Kalomeris, Jacobs, & Hiller, 2017). While glucose and lactate were well controlled (Figure 7b and 7c), ammonium accumulated to a maximum of 5.5, 8.9, and 15.5 mM during the steady state conditions for the 1:2.5, 1:5, and 1:10 volume ratios CSTRs, respectively. It is possible that the production CSTRs were slightly overfed and the cascading CSTRs process could be improved by optimizing the nutrient feed rate.

The steady state volumetric productivities for the dual cascading CSTRs were calculated using the same methods as described for the N-1 perfusion linked to production CSTRs

via the hydrocyclone. The instantaneous product concentration and total product produced are shown in Figures 7d and 7e, respectively; Table 3 summarizes the steady state values. The cascading CSTR systems reached volumetric productivities of 0.27 - 0.53 g/L/day. Although the operation of a linked bioreactor system using a hydrocyclone is more complex than the cascading CSTR system, it also achieves 1.8 - 3.1 times the productivity. At a volumetric productivity of 0.53 g/L/day, the unoptimized cascading CSTR system with a 1:2.5 N-1 to production bioreactor volume ratio had a slightly higher volumetric productivity than the 12-day optimized fed-batch process and could be a simple alternative when a continuous process is desirable.

To further optimize the process, the dilution rate in the N-1 CSTR might be varied. However, the maximum dilution rate is limited by washout conditions. At dilution rates approaching the washout dilution rate, the cell density would be low and result in less cells overall being delivered to the production CSTR. At dilution rates significantly below the washout dilution rate, the cells in the N-1 CSTR would grow at a slower rate which would make them less likely to continue to divide upon reaching the production CSTR. Cell retention in the N-1 bioreactor resolves these issues with high perfusion rates and the corresponding high growth rates which allow for large quantities of cells to be delivered to the production CSTR. The volume ratio between the N-1 and production CSTR could also be optimized. It is possible that a higher volumetric productivity could be attained if the volume of the N-1 CSTR is increased relative to the size of the production CSTR. However, this may not be practical if the intention is to make use of existing manufacturing infrastructure.

Once the cell bleed addition from the N-1 was halted on Day 35 in the 1:2.5 and 1:5 simulated volume ratio production CSTRs, a drop in viable cell density is observed (Figures 6b and 6c). The product concentration in the bioreactor also declined (Figure 7d), indicating a drop in volumetric productivity in the independently operating production CSTRs. While the experiment did not achieve a second steady state for these two bioreactors, and a complete dilution rate study was not attempted, it is clear from the limited data that the cascading linked bioreactors system is significantly more productive than the stand alone CSTRs for similar operating conditions. Although only modest productivities were achieved while operating as a stand-alone CSTR without cell retention in the current work, recent public disclosures on standalone CSTRs for cell culture (Khattak et al., 2019) demonstrate the potential for significant improvement with optimization.

4. CONCLUSIONS

We have demonstrated a highly productive and media efficient continuous linked bioreactor system using a hydrocyclone as a cell retention device. This process has the potential to reduce manufacturing costs and increase volumetric productivity in existing manufacturing facilities while eliminating concerns of membrane fouling. In a manufacturing setting, a single hydrocyclone of the design utilized in this work could operate as the cell retention device on a highly proliferative 500-L N-1 perfusion while supplying cells to a 2,500-10,000-L production CSTR. This system would operate at a steady state volumetric productivity as high as 0.96 g/L/day, require only 0.17 RV/day of media, and deliver a continuous stream of antibody product at 5.7 grams/L. These volumetric productivities are comparable to a linked bioreactor system using a membrane

cell retention device (Gagnon et al., 2019; Hiller et al., 2019) and up to twice as productive as the optimized fed-batch process. Perfusion with the hydrocyclone would have significant advantages if applied to large scale bioreactors for which conventional membrane perfusion methods may be impractical. Although we found that the hydrocyclone had operational limitations for other modes of perfusion, it was successfully used for a highly productive linked bioreactor system that could be adapted for legacy manufacturing infrastructure without extraordinary facility modification, enabling the economic benefits of a continuous or semi-continuous downstream purification process.

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AUTHOR CONTRIBUTION STATEMENT

Gregory Hiller conceptualized the idea of using a hydrocyclone as a cell retention device for the linked bioreactor system. He identified practical volume ratios between the bioreactors as well as dilution rates that may yield the highest volumetric productivities. Anita Kundu realized that the cascading CSTR system should also be compared to the linked bioreactor system using a hydrocyclone. She designed the 3D printed

hydrocyclone, performed the bioreactor experiments, and analyzed the data. Both authors contributed to drafting the manuscript.

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TABLES

Table 1. Compositions of media and feeds

Medium	Total Amino Acids (mM)	Glucose (g/L)	Osmolality (mOsm/kg)	Poly-vinyl alcohol (g/L)	pH
Basal medium	120	4	280	2.6	7.1
N-1 perfusion/dilution medium	90	5	320	5.1	7.2
Production CSTR concentrated feed	600	50	1080	5.1	7.2

Table 2. Summary of N-1 perfusion bioreactor linked to a production CSTR steady states

Bioreactor	Steady State	Days	Average VCD ($\times 10^6$ cells/mL)	Average Perfusion or Dilution Rate (RV/day)	Volumetric Productivity (g/L/day) calculated by linear regression	Volumetric Productivity (g/L/day) calculated by multiplying titer by dilution or perfusion rate	Volume of Culture Received from N-1 Perfusion (RV/day)	Average Volume of Concentrated Nutrient Feeds (RV/day) †	Mass of Product per Volume of Liquid Used (g/L)	Average Steady State Cell Specific Productivity (pg/cell/day)
N-1 Perfusion (operating independently)	1	18-23	20.4	1.05	0.54	0.51	N/A	N/A	0.51	26.6
	2	27-33	16.5	1.40	0.36	0.35	N/A	N/A	0.26	21.9
1:5 Simulated Volume Ratio CSTR	1	18-21	19.0	0.27	0.93	0.84	0.21	0.05	3.44	42.5
	2	28-33	15.0	0.32	0.55	0.58	0.28	0.03	1.72	31.5
1:10 Simulated Volume Ratio CSTR	1	18-21	23.5	0.17	0.96	0.68	0.11	0.06	5.65	38.0
	2	27-33	14.9	0.18	0.54	0.63	0.14	0.04	3.00	33.0
1:20 Simulated Volume Ratio CSTR	1	19-23	20.9	0.10	0.84	0.47	0.05	0.05	8.40	39.5
	2	31-33	11.5	0.10	0.54	0.55	0.07	0.02	5.40	45.6

† In addition to concentrated nutrient feeds, small volumes of 1M sodium/potassium carbonate and 1% simethicone were added to the bioreactors to control pH and foaming, respectively.

Table 3. Summary of Cascading CSTRs Steady States

Bioreactor	Days	Average VCD ($\times 10^6$ cells/mL)	Average Dilution Rate (RV/day)	Volumetric Productivity (g/L/day) calculated by linear regression	Volumetric Productivity (g/L/day) calculated by multiplying titer by dilution rate	Average Steady State Cell Specific Productivity (pg/cell/day)
N-1 CSTR (operating independently)	14-43	10.0	0.53	0.14	0.11	20.1
1:2.5 Simulated Volume Ratio CSTR	22-35	16.1	0.25	0.53	0.51	29.7
1:5 Simulated Volume Ratio CSTR	24-35	12.9	0.14	0.43	0.44	31.4
1:10 Simulated Volume Ratio CSTR	39-43	6.0	0.07	0.27	0.33	44.1

FIGURES

Figure 1. A schematic of the hydrocyclone. The inlet stream enters the top of the device from two ports directing the flow tangential to the inner conical chamber. An internal centrifugal force is generated creating a primary vortex. Cells are pushed outward towards the wall of the swirl chamber and fluid is directed downward by the conical taper until the concentrated cells exit in the underflow stream through the bottom port. Simultaneously, partially cell-depleted fluid reverses direction moving upward in a secondary vortex. This stream (the overflow) exits through a port at the top.

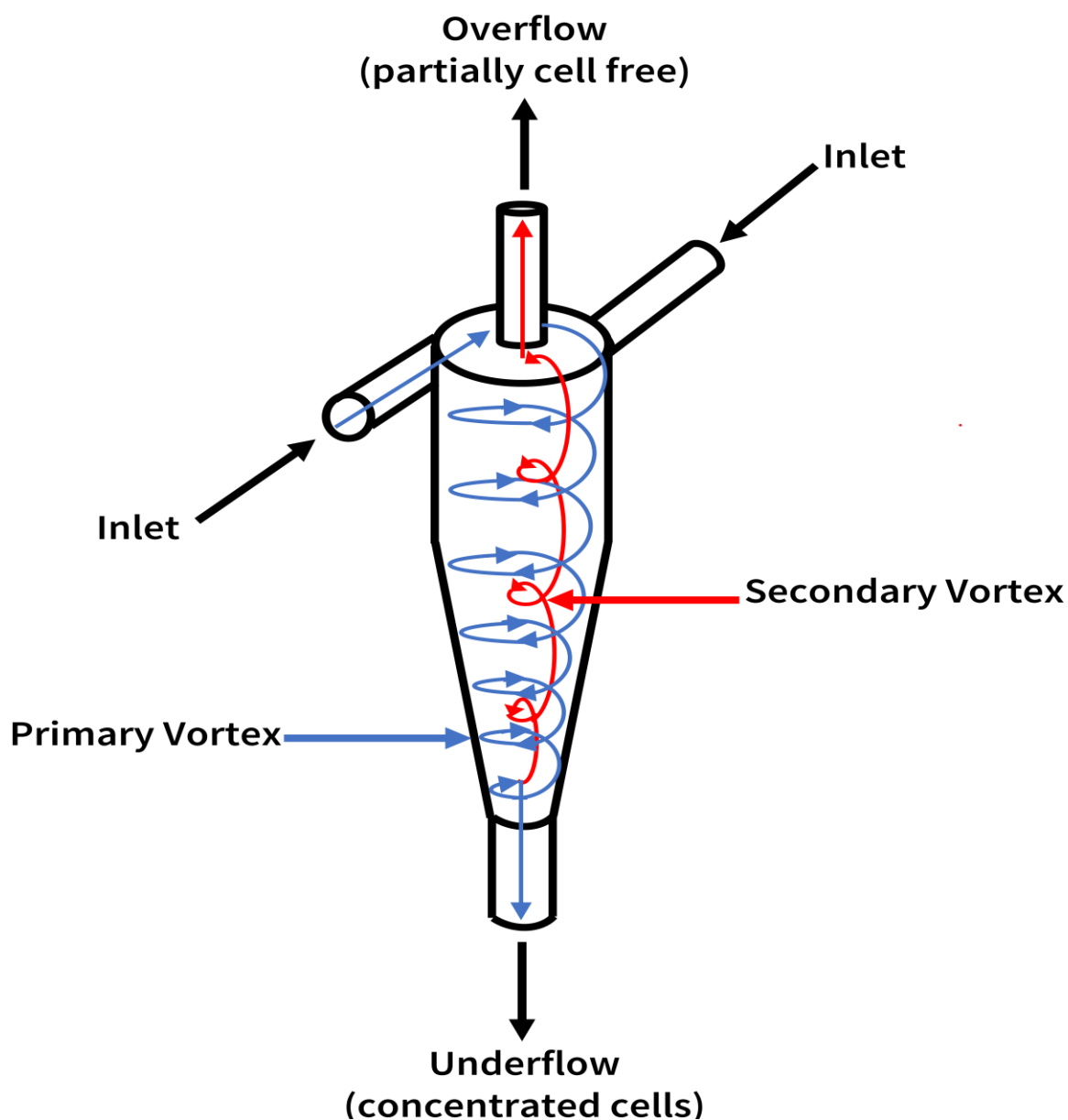


Figure 2. Schematics of the linked bioreactor system using a hydrocyclone and cascading CSTRs system. **a)** An N-1 perfusion bioreactor operates at steady state with a hydrocyclone cell retention device. A peristaltic pump supplies flow to the hydrocyclone. The culture exits the hydrocyclone in two outlet streams: an underflow stream consisting of concentrated cells that is recirculated back to the N-1 perfusion bioreactor, and a partially cell-depleted overflow stream. A second peristaltic pump controls the flowrate of the overflow stream which provides a continuous inoculum source for the larger production CSTR. The production CSTR is fed with concentrated nutrient feeds and is maintained at constant volume by continuously removing whole cell harvest. **b)** This process does not require any cell retention. The N-1 CSTR is operated at a high dilution rate with the cell bleed continuously feeding a larger production CSTR. In addition to residual nutrients entering with cells from the N-1, the production CSTR is fed directly with a concentrated nutrient feed and maintained at constant volume by continuously removing whole cell harvest.

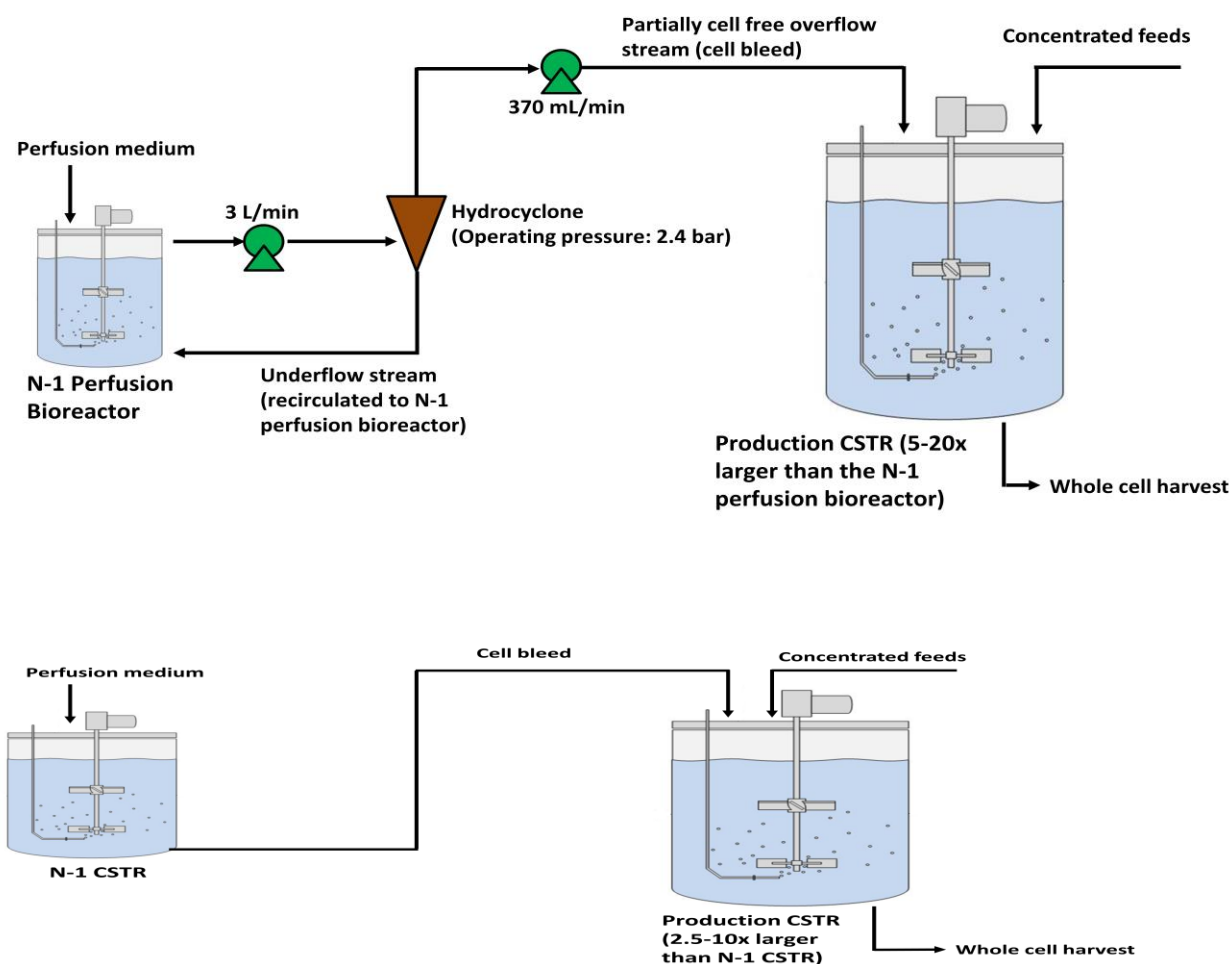


Figure 3. Hydrocyclone and N-1 perfusion bioreactor performance data including **a)** total separation efficiency and reduced separation efficiency achieved by the hydrocyclone, and **b)** perfusion rate and calculated effective cell bleed in the N-1 perfusion bioreactor (the dashed vertical lines indicate the bounds of a steady state).

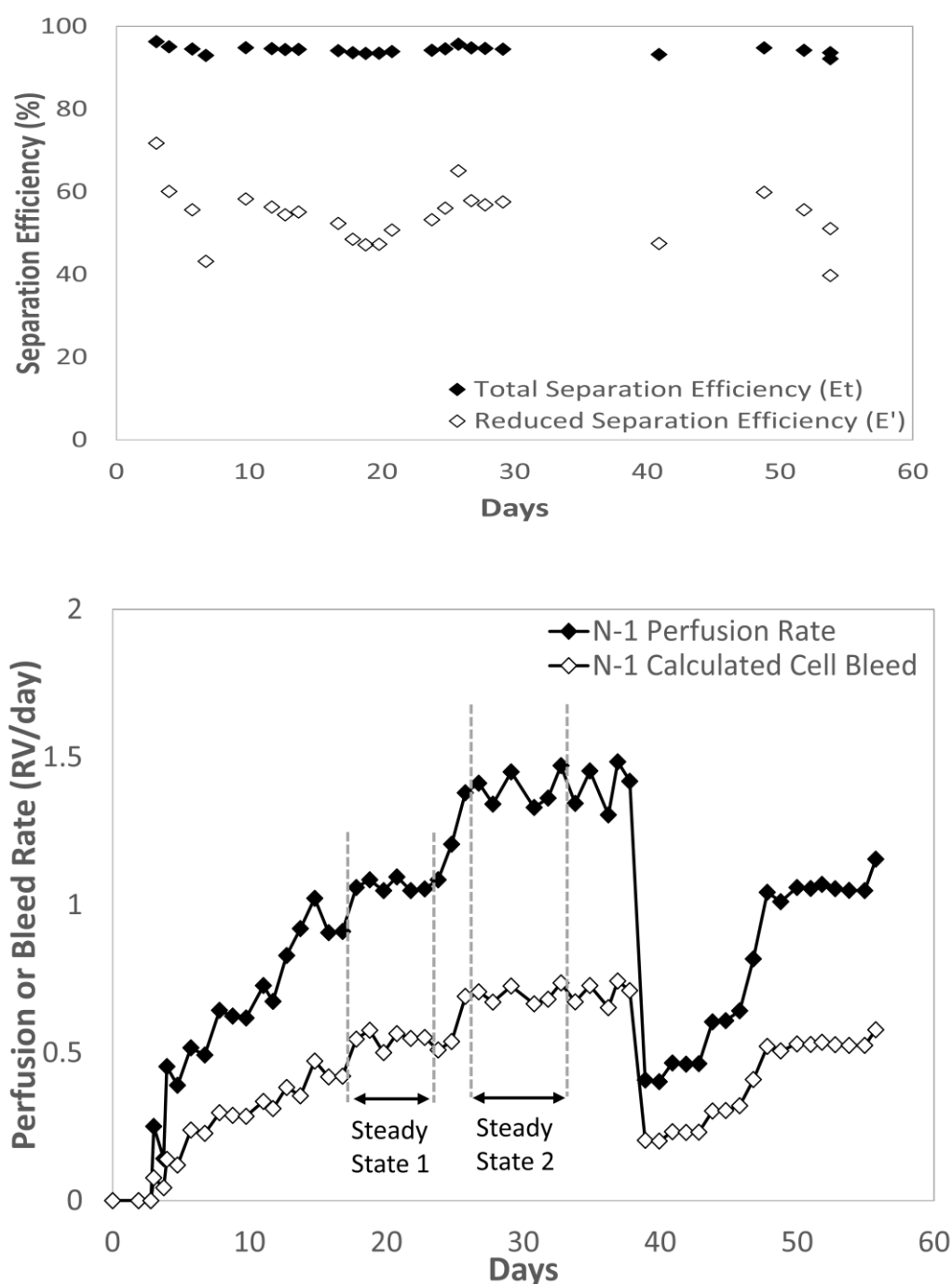
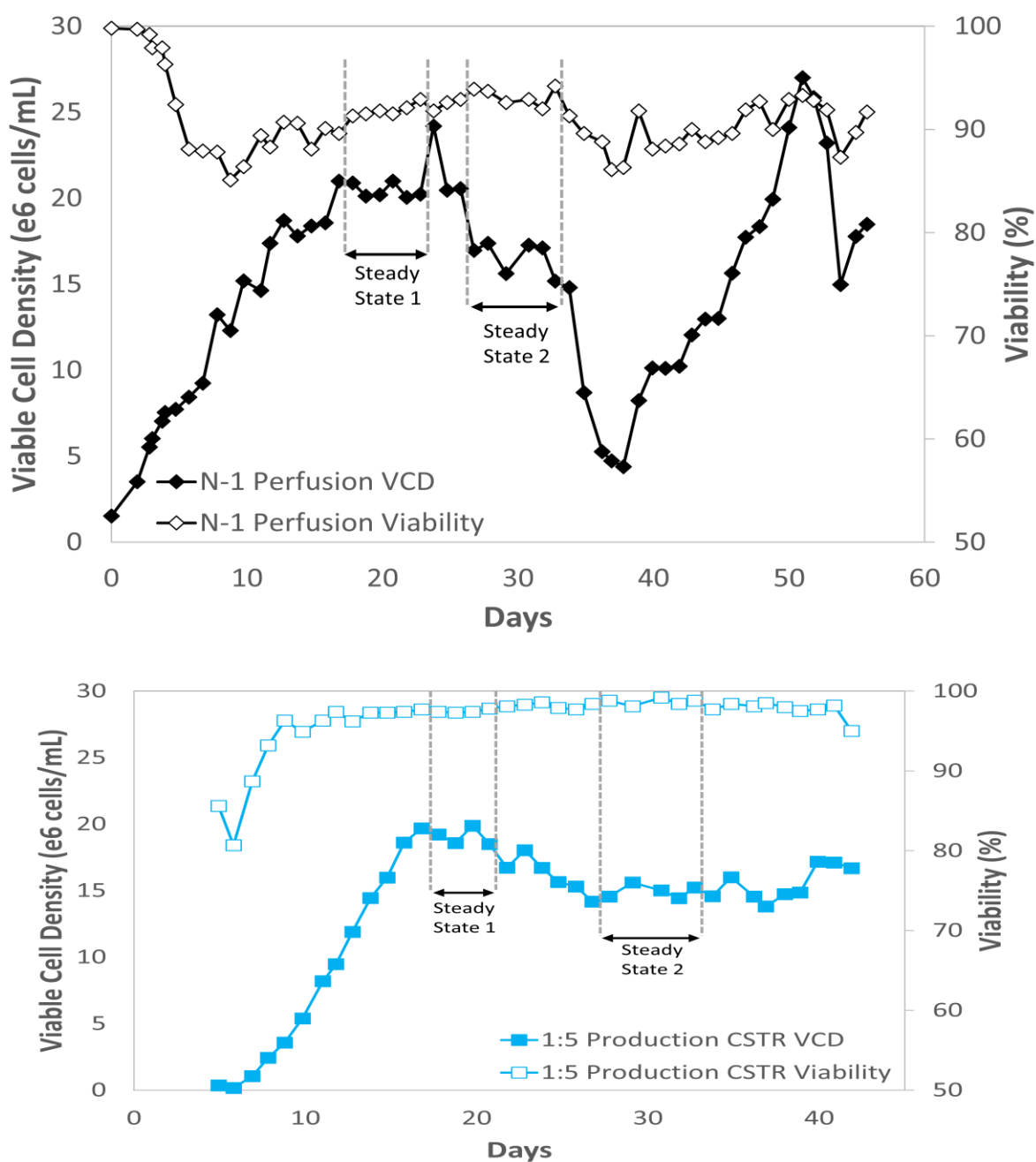


Figure 4. Viable cell density (VCD) and cell viability in the linked bioreactor system using a hydrocyclone for **a)** N-1 perfusion bioreactor, **b)** 1:5 simulated volume ratio production CSTR, **c)** 1:10 simulated ratio production CSTR, and **d)** 1:20 simulated ratio production CSTR. The dashed lines denote the bounds of a steady state.



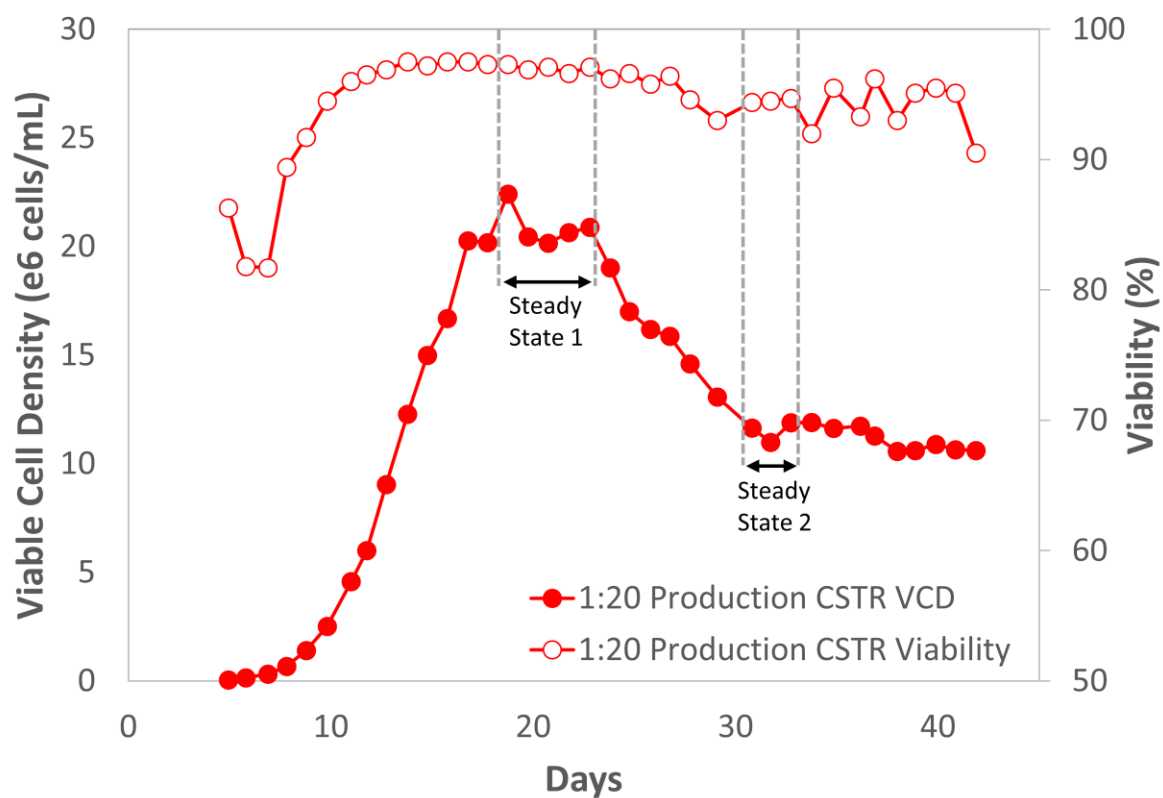
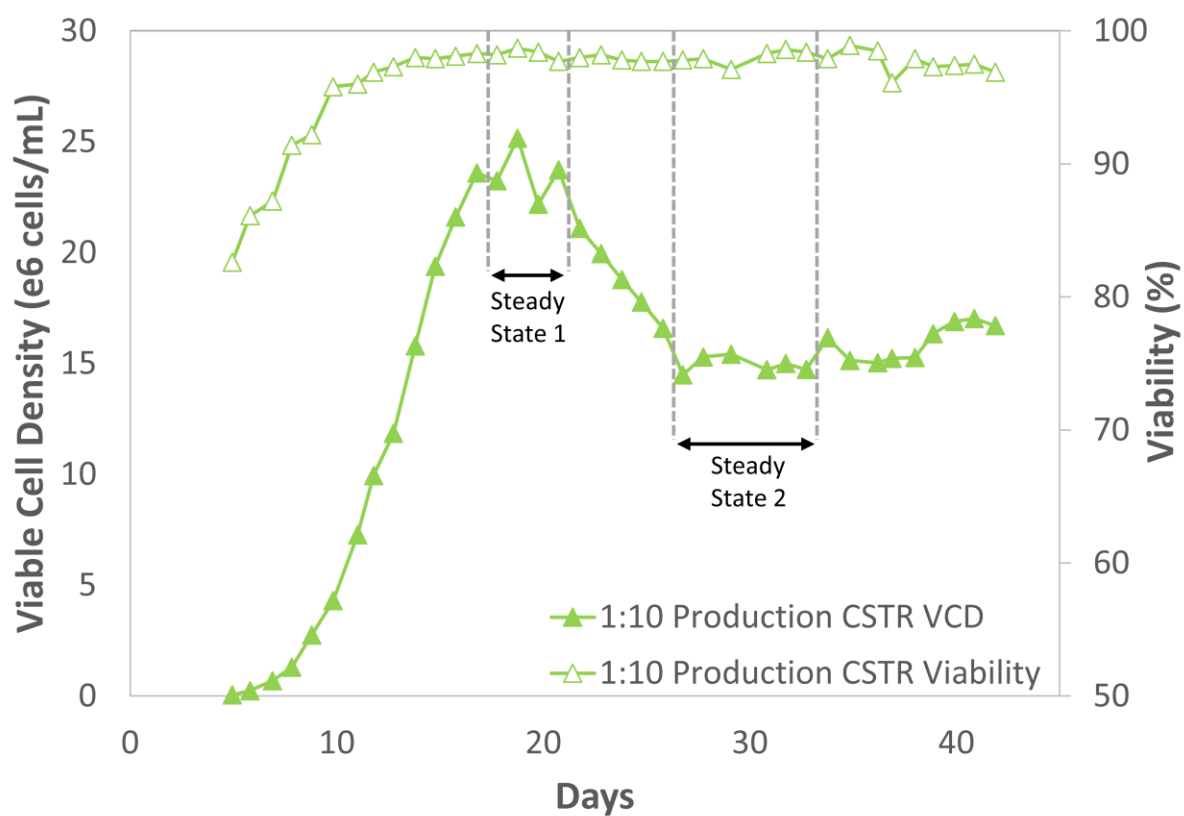
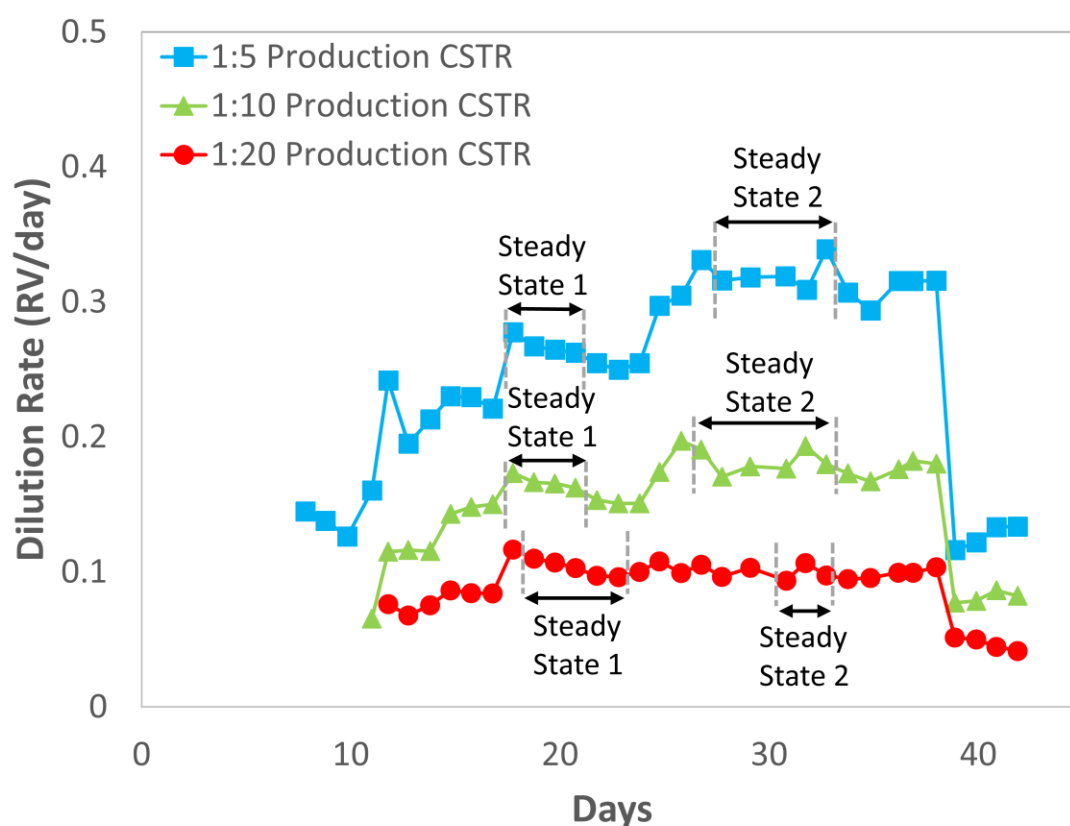
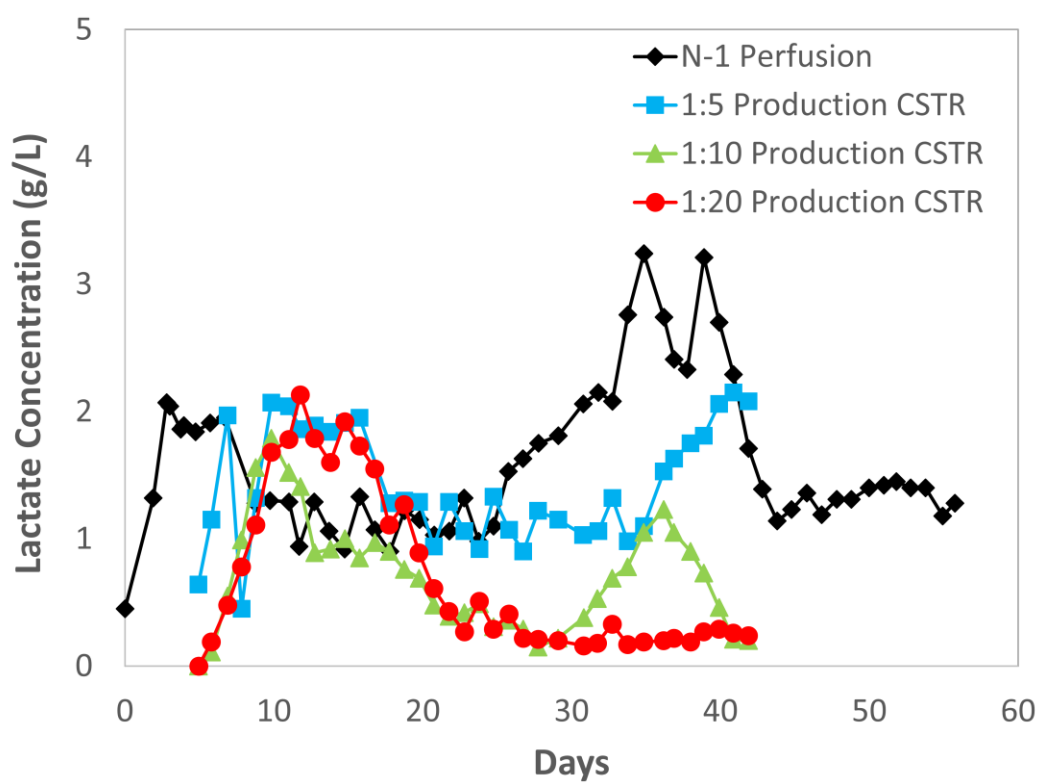
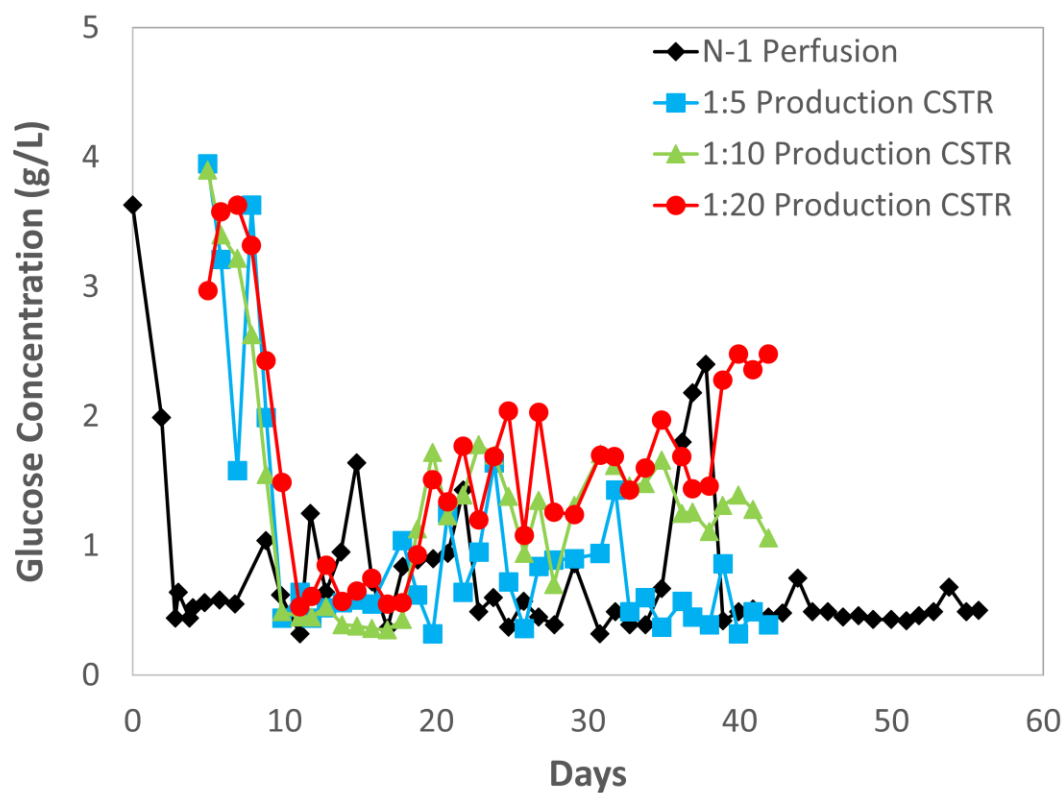


Figure 5. Bioreactor cell culture performance data for the linked bioreactor system including **a)** dilution rates in the production CSTRs (the dashed lines denote the bounds of the steady states), **b)** residual glucose concentration, **c)** lactate concentration, **d)** instantaneous product concentration or titer, and **e)** total product produced in the linked bioreactor system divided by the bioreactor working volume during the two steady states (circled); a linear regression was performed during the steady state timepoints to determine the steady state volumetric productivity (equal to the slope of the regression).





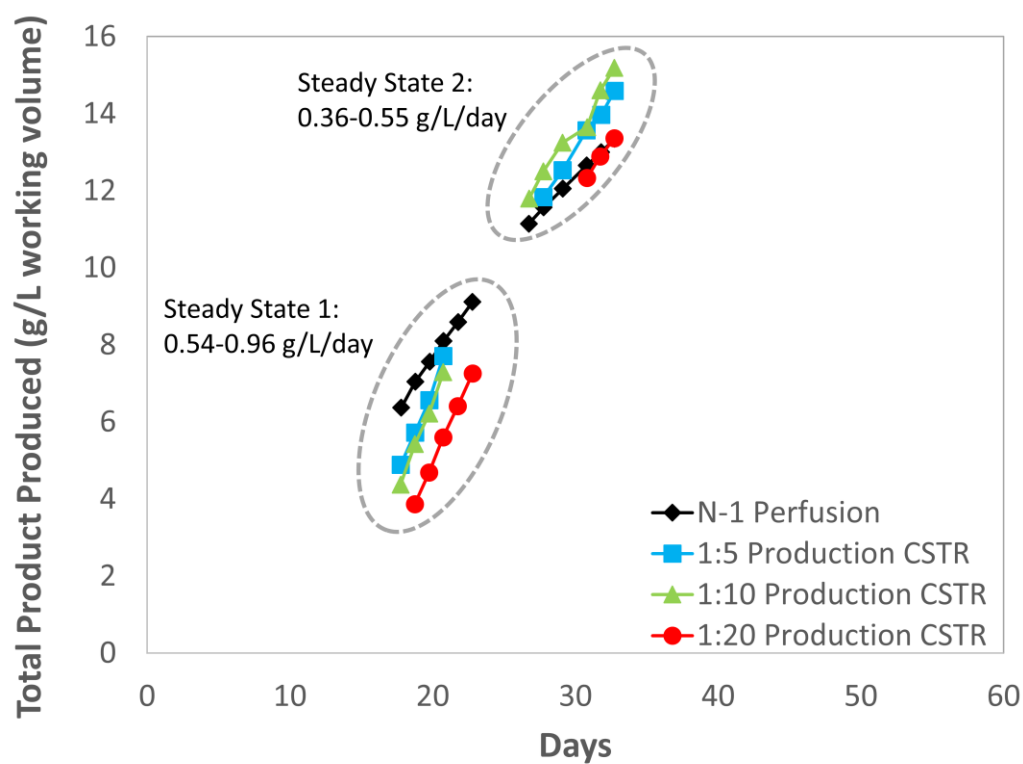
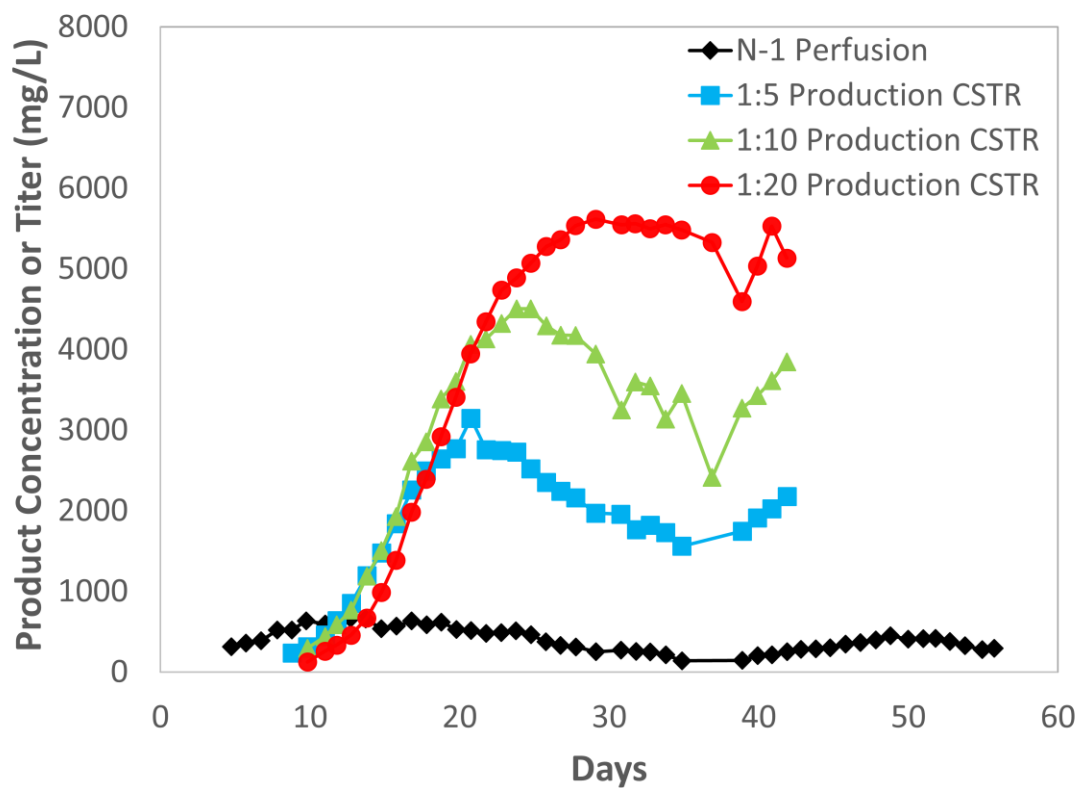
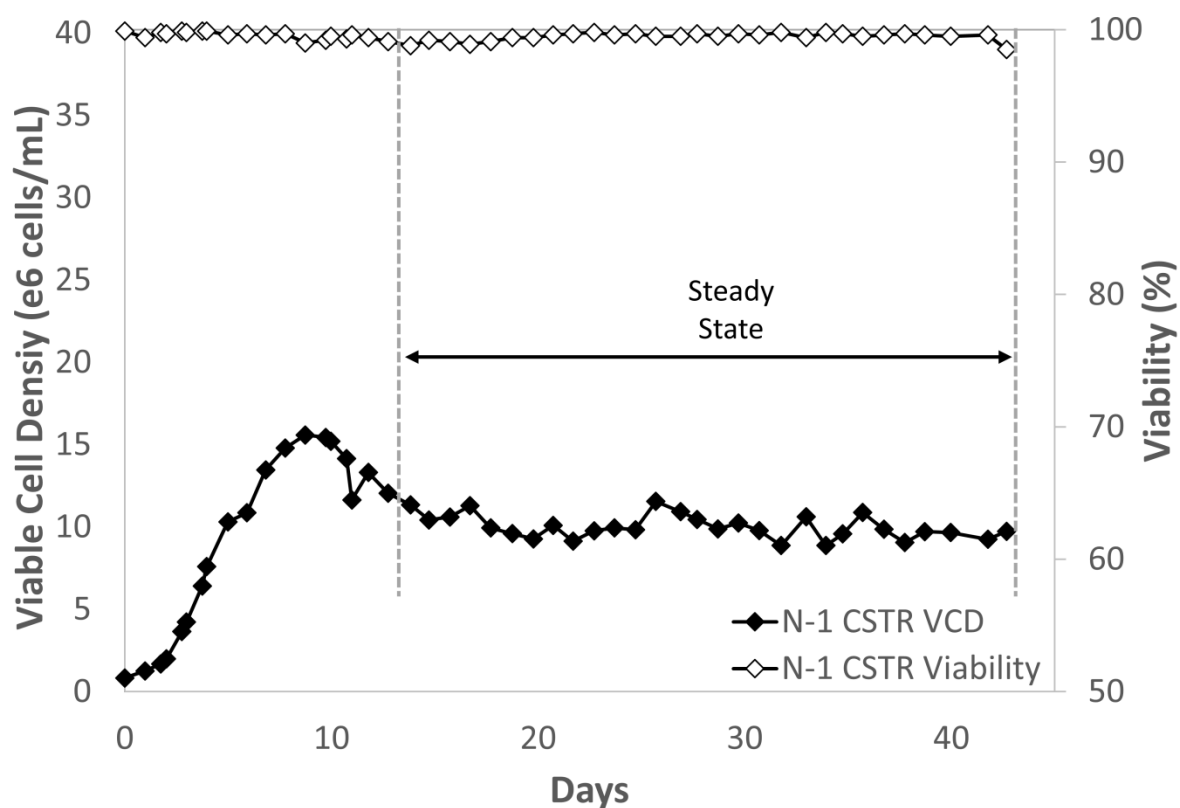
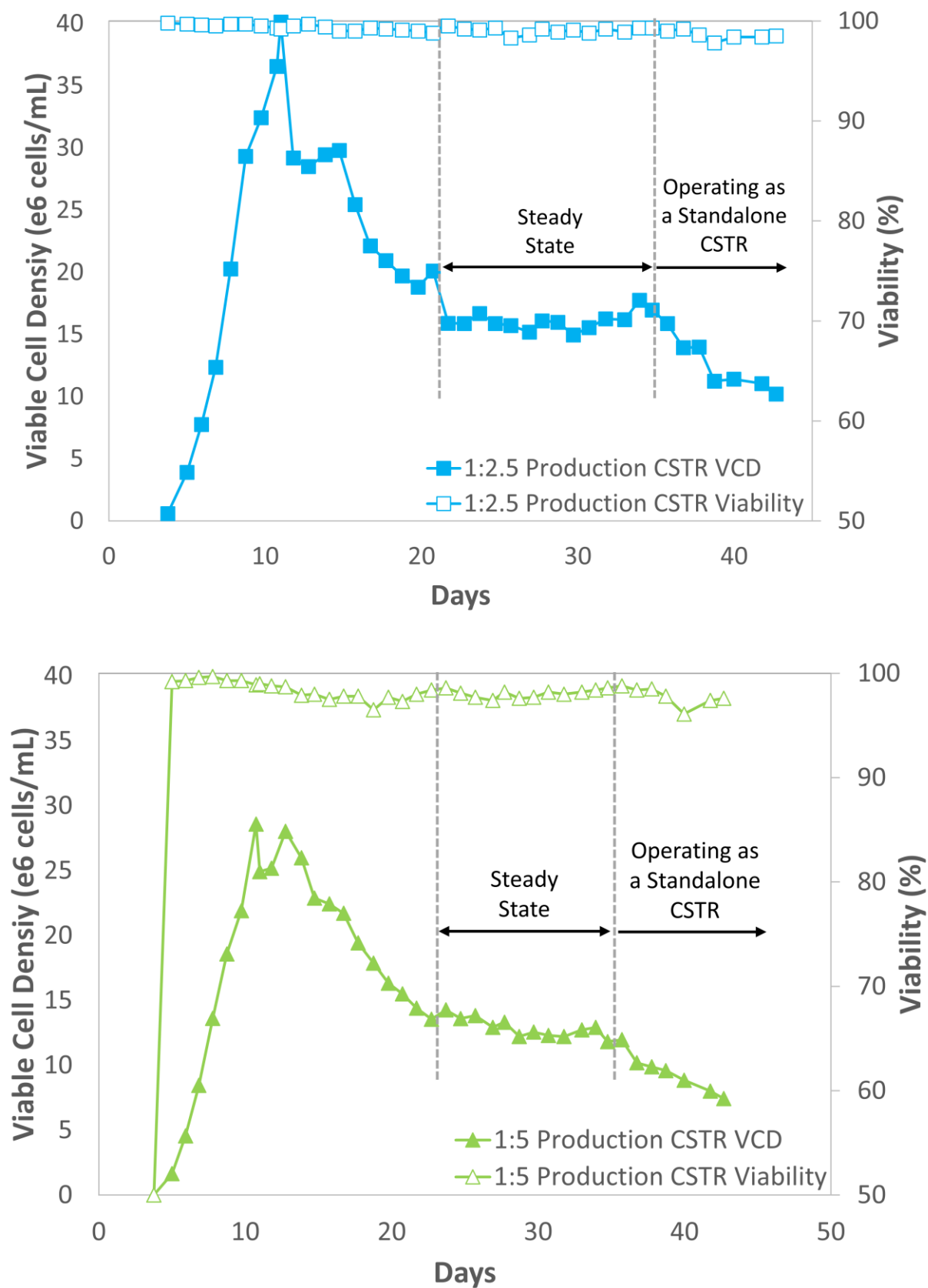


Figure 6. Viable cell density (VCD) and cell viability profiles in the cascading CSTRs system for **a)** N-1 CSTR, **b)** 1:2.5 simulated volume ratio production CSTR, **c)** 1:5 simulated volume ratio production CSTR, and **d)** 1:10 simulated volume ratio production CSTR. The dashed lines denote the bounds of the steady states and the timepoint at which the CSTR began operating as an independent unit operation.





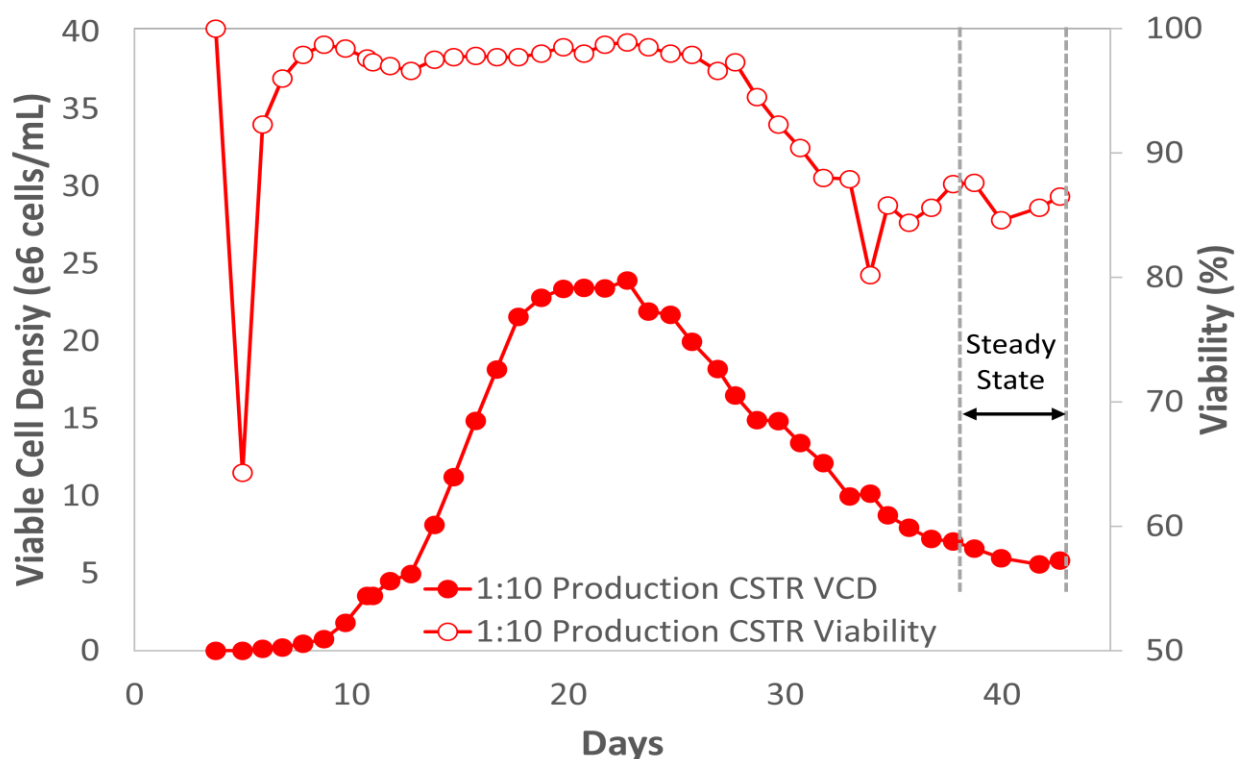


Figure 7. Bioreactor cell culture performance data for the cascading CSTRs system including **a)** dilution rates (the dashed lines denote the bounds of the steady states), **b)** residual glucose concentration, **c)** lactate concentration, **d)** instantaneous product concentration or titer, **e)** total product produced in the cascading CSTR system divided by the bioreactor working volume during the steady states; a linear regression was performed during the steady state timepoints to determine the steady state volumetric productivity (equal to the slope of the regression).

